INCH-POUND

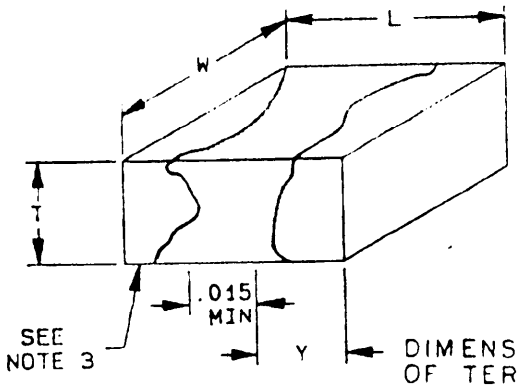
MIL-C-123/14
21 November 1989

MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY, NONLEADED (HIGH FREQUENCY), STYLE CKS28

This specification is approved for use by all Departments and Agencies of the Department of Defense.

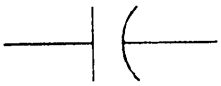
The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-123.




DIMENSION MAY GO TO .005 AT EDGES OF TERMINATIONS

| Inches | mm |
|---|---|
| .005 | 0.13 |
| .011 | 0.28 |
| .012 | 0.30 |
| .015 | 0.38 |
| .016 | 0.40 |
| .020 | 0.51 |
| .055 | 1.40 |
| .059 | 1.50 |

| Dimensions | | | | |
|---|---|---|---|---|
| | | T | | |
| L ±.012 | W ±.012 | Min. | Max. | Y ±.011 |
| .055 | .055 | .020 | .059 | .016 |



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions are for bare chips; for solder coated terminations add .015 to the length and .012 to the width and thickness envelope.

FIGURE 1. Style CKS28 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Style CKS28 characteristics.

| Part number 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (V dc) |
|---|---|---|---|---|
| M123A14B-C0R1B- | 0.1 | B | BG, BP | 150 |
| M123A14B-C0R2B- | 0.2 | B | " | " |
| M123A14B-C0R3-- | 0.3 | B, C | " | " |
| M123A14B-C0R4-- | 0.4 | " | " | " |
| M123A14B-C0R5-- | 0.5 | " | " | " |
| M123A14B-C0R6-- | 0.6 | B, C, D | " | " |
| M123A14B-C0R7-- | 0.7 | " | " | " |
| M123A14B-C0R8-- | 0.8 | " | " | " |
| M123A14B-C0R9-- | 0.9 | " | " | " |
| M123A14B-C1R0-- | 1.0 | " | " | " |
| M123A14B-C1R1-- | 1.1 | " | " | " |
| M123A14B-C1R2-- | 1.2 | " | " | " |
| M123A14B-C1R3-- | 1.3 | " | " | " |
| M123A14B-C1R4-- | 1.4 | " | " | " |
| M123A14B-C1R5-- | 1.5 | " | " | " |
| M123A14B-C1R6-- | 1.6 | " | " | " |
| M123A14B-C1R7-- | 1.7 | " | " | " |
| M123A14B-C1R8-- | 1.8 | " | " | " |
| M123A14B-C1R9-- | 1.9 | " | " | " |
| M123A14B-C2R0-- | 2.0 | " | " | " |
| M123A14B-C2R1-- | 2.1 | " | " | " |
| M123A14B-C2R2-- | 2.2 | " | " | " |
| M123A14B-C2R4-- | 2.4 | " | " | " |
| M123A14B-C2R7-- | 2.7 | " | " | " |
| M123A14B-C3R0-- | 3.0 | " | " | " |
| M123A14B-C3R3-- | 3.3 | " | " | " |
| M123A14B-C3R6-- | 3.6 | " | " | " |
| M123A14B-C3R9-- | 3.9 | " | " | " |
| M123A14B-C4R3-- | 4.3 | " | " | " |
| M123A14B-C4R7-- | 4.7 | " | " | " |
| M123A14B-C5R1-- | 5.1 | " | " | " |
| M123A14B-C5R6-- | 5.6 | " | " | " |
| M123A14B-C6R2-- | 6.2 | " | " | " |
| M123A14B-C6R8-- | 6.8 | B, C, J, K | " | " |
| M123A14B-C7R5-- | 7.5 | " | " | " |
| M123A14B-C8R2-- | 8.2 | " | " | " |
| M123A14B-C9R1-- | 9.1 | " | " | " |
| M123A14B-C100-- | 10 | F, G, J, K | " | " |
| M123A14B-C110-- | 11 | " | " | " |
| M123A14B-C120-- | 12 | " | " | " |
| M123A14B-C130-- | 13 | " | " | " |
| M123A14B-C150-- | 15 | " | " | " |
| M123A14B-C160-- | 16 | " | " | " |
| M123A14B-C180-- | 18 | " | " | " |
| M123A14B-C200-- | 20 | " | " | " |
| M123A14B-C220-- | 22 | " | " | " |
| M123A14B-C240-- | 24 | " | " | " |
| M123A14B-C270-- | 27 | " | " | " |
| M123A14B-C300-- | 30 | " | " | " |
| M123A14B-C330-- | 33 | " | " | " |
| M123A14B-C360-- | 36 | " | " | " |

See footnote at end of table.



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TABLE I. Style CKS28 characteristics - Continued.

| Part number 1/ | Capacitance (pF) | Capacitance tolerance | Voltage-temperature limits | Rated voltage (V dc) |
|---|---|---|---|---|
| M123A14B-C390-- | 39 | F, G, J, K | BG, BP | 150 |
| M123A14B-C430-- | 43 | F, G, J, K | " | " |
| M123A14B-C470-- | 47 | F, G, J, K, M | " | " |
| M123A14B-C510-- | 51 | " | " | " |
| M123A14B-C560-- | 56 | " | " | " |
| M123A14B-C620-- | 62 | " | " | " |
| M123A14B-C680-- | 68 | " | " | " |
| M123A14B-C750-- | 75 | " | " | " |
| M123A14B-C820-- | 82 | " | " | " |
| M123A14B-C910-- | 91 | " | " | " |
| M123A14B-C101-- | 100 | " | " | " |

1/ The complete part number shall include additional symbols to indicate characteristic, capacitance tolerance, and termination, as applicable.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multi-layer, unencapsulated, monolithic.

Material: Ceramic.

Terminations: In accordance with MIL-C-123; symbols G, S, and T only.

Inspection conditions: All visual examinations shall be performed at 20X magnification, using a binocular microscope.

Capacitance: See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I.

Operating temperature: -55°C to +125°C.

Dissipation factor: 0.025 percent maximum at 1 MHz.

Insulation resistance:

At +25°C ±5°C: $10^6$ megohms, minimum.

At +125°C ±5°C: $10^5$ megohms, minimum.

Equivalent series resistance: In accordance with MIL-C-123.

Resonance frequency: In accordance with MIL-C-123.

Application note: These capacitors are intended for use in RF solid-state circuitry up to and including 25 GHz and cannot be replaced in the field.



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CONCLUDING MATERIAL

Custodians:
Air Force - 19
NASA - NA

Review activities:
Air Force - 85
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5910-1574-1)



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